

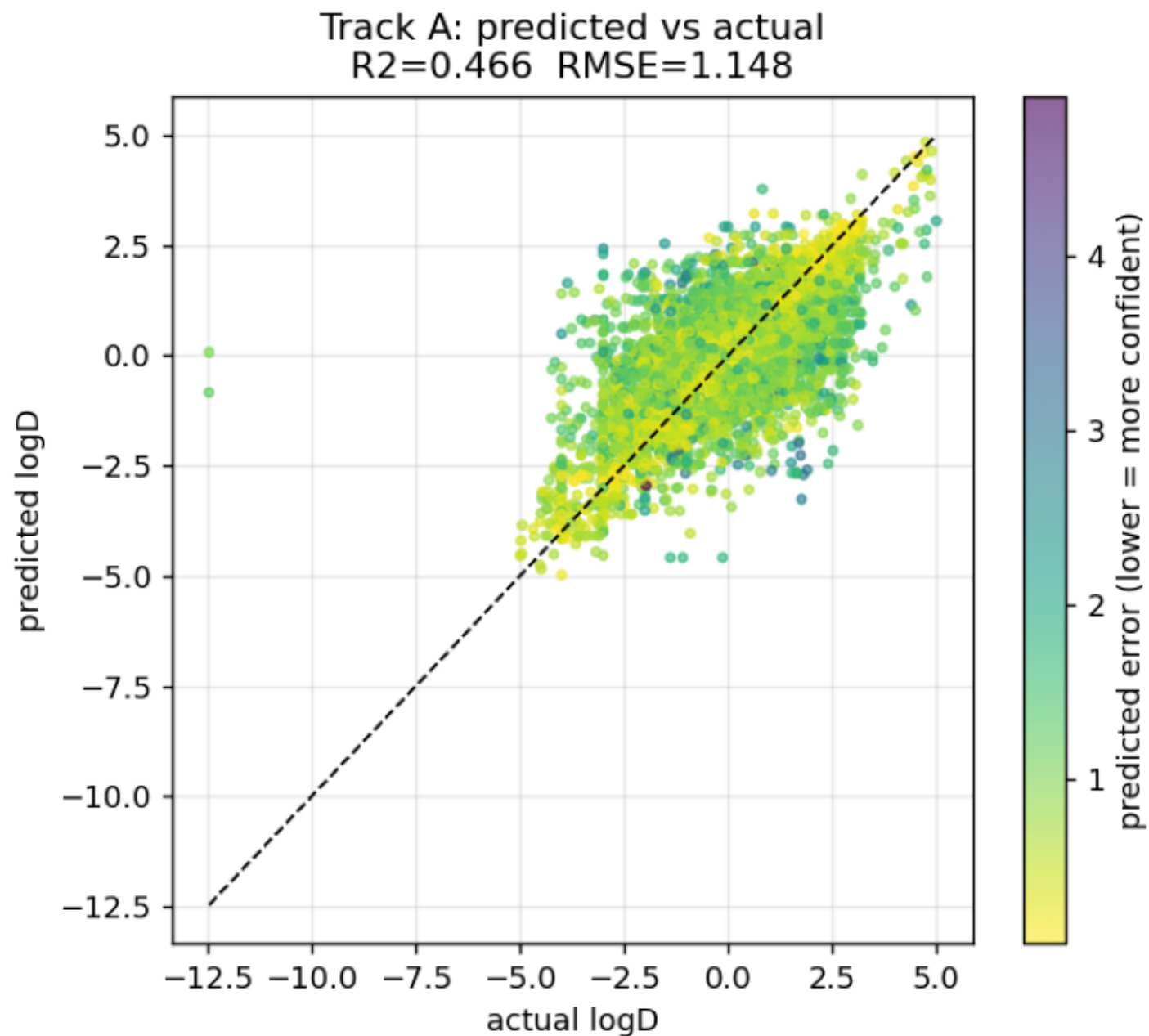


Figure 1. Track A, new-extractant screening: predicted vs actual logD under molecule-grouped cross-validation. Overall R^2 0.466 (RMSE 1.148); on the most confident 10 percent, R^2 0.912 (RMSE 0.493).

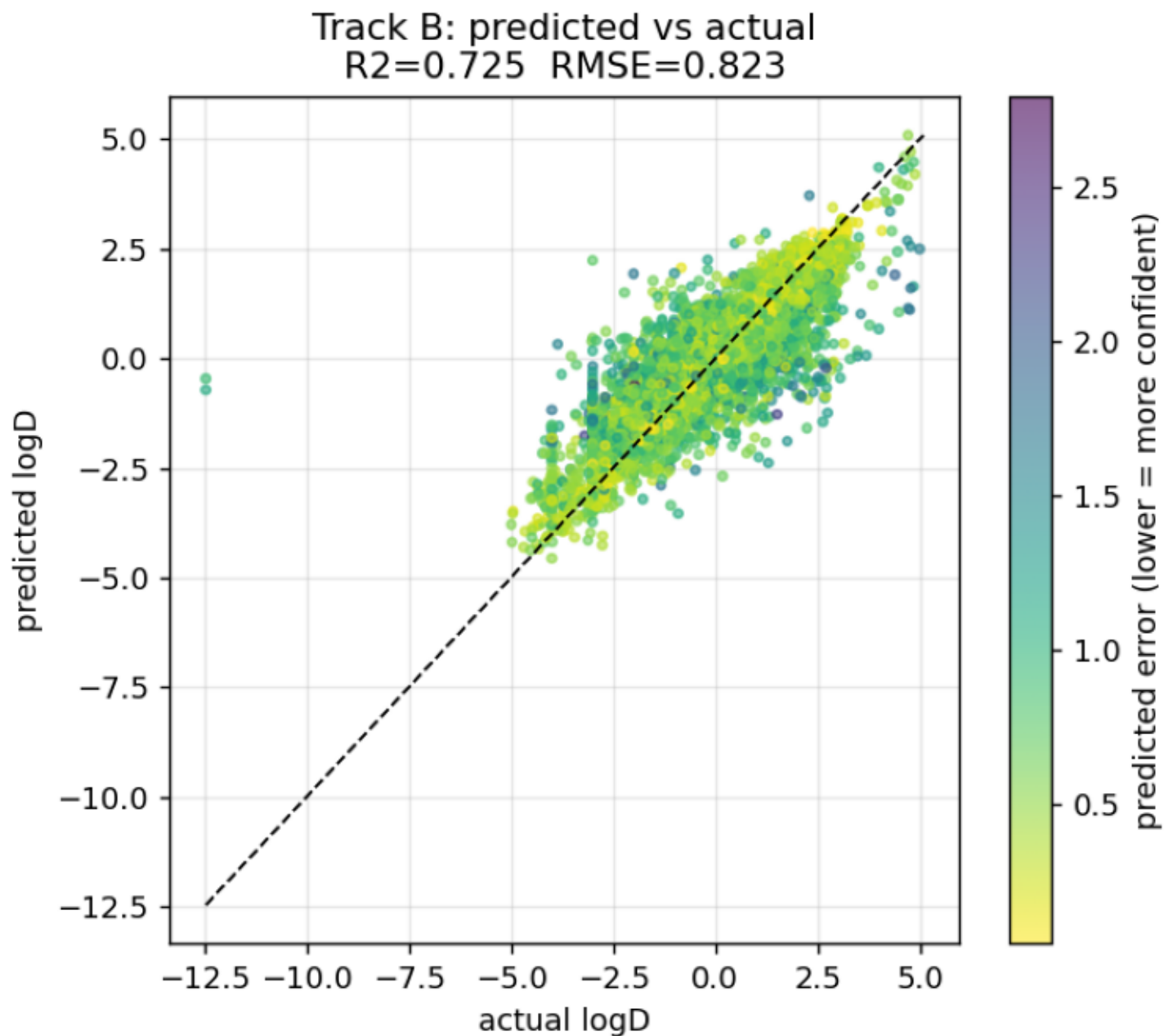


Figure 2. Track B, condition optimization for a known extractant: predicted vs actual logD. Honest condition-key R^2 0.61 (RMSE 0.98); most confident 10 percent R^2 0.874 (RMSE 0.430).

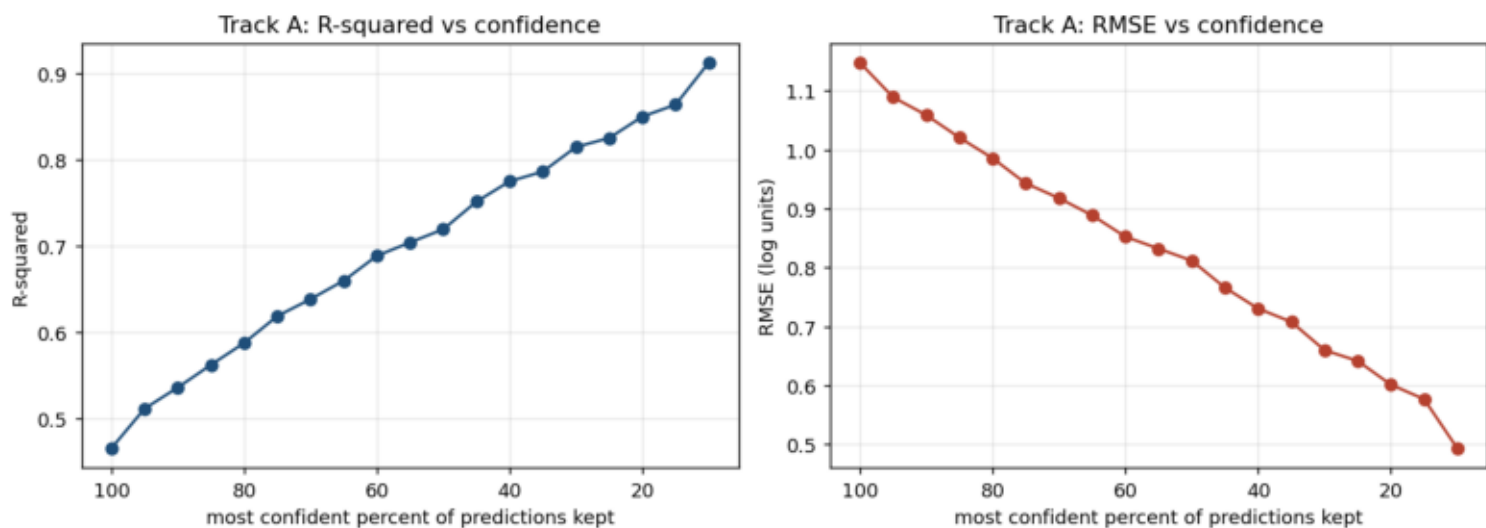


Figure 3. Accuracy concentrates with confidence (Track A): error falls and R2 rises as predictions are restricted to the most confident fraction. The calibrated confidence layer is the model's deployable contribution.

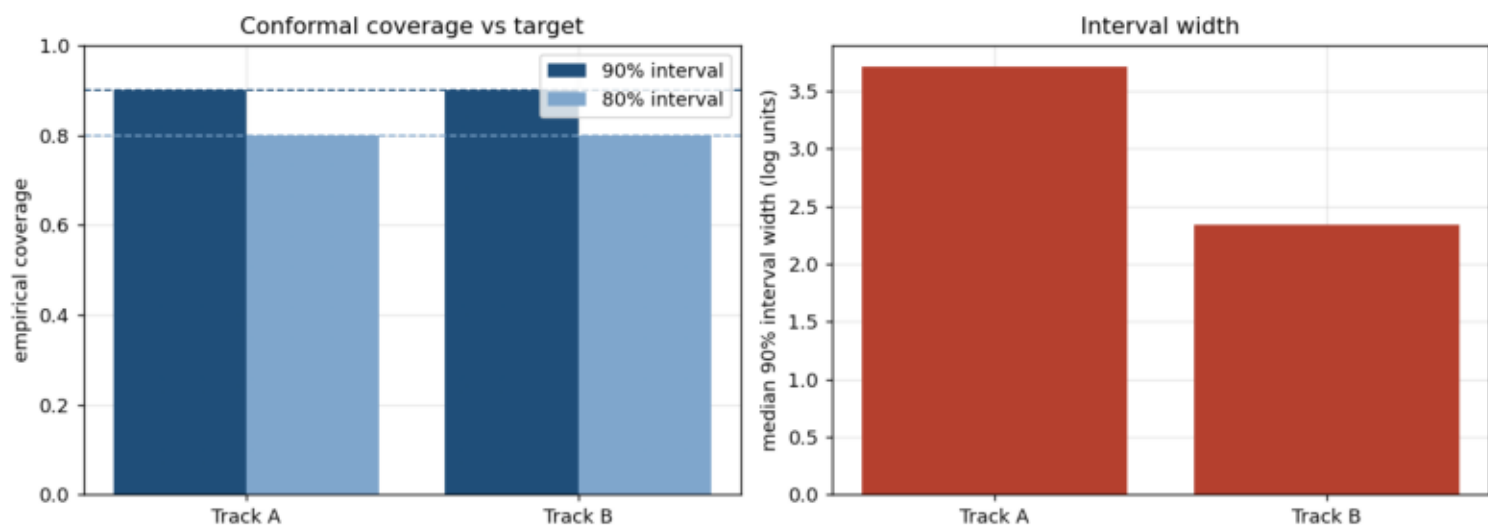


Figure 4. Calibrated 90 percent prediction intervals: empirical coverage 0.902 on a molecule-disjoint split, so the uncertainty is trustworthy rather than nominal.

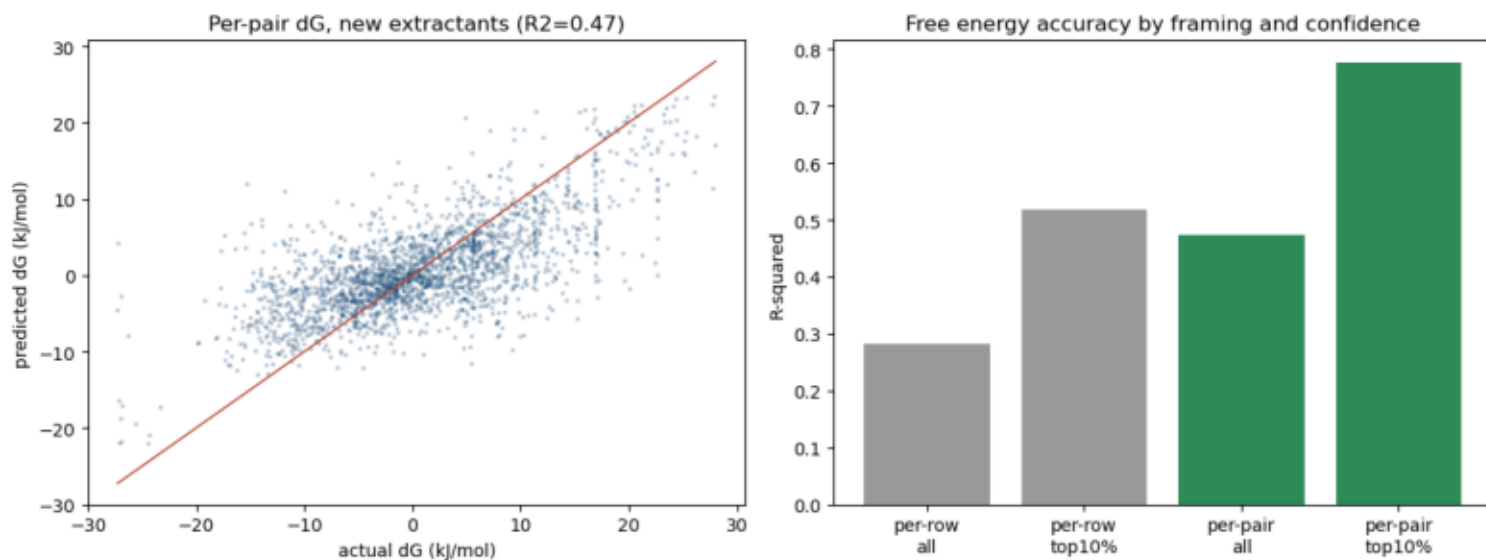


Figure 5. Free energy of extraction (delta G), per-pair framing: R2 0.46 plus or minus 0.01 (RMSE 6.3 kJ/mol); most confident 10 percent RMSE 3.6 kJ/mol. RandomForest chosen by a seven-model bake-off.

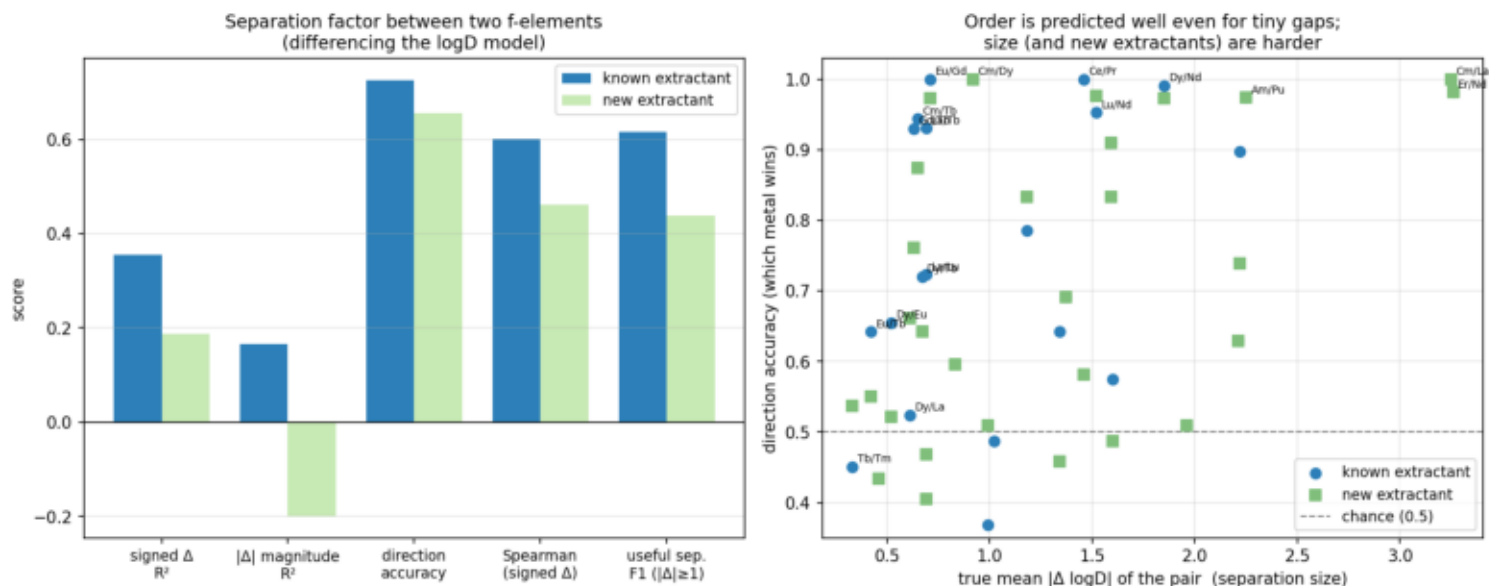


Figure 6. Separation factor between two f-elements (the objective), differencing the logD model: direction is predicted moderately (0.73 known, 0.66 new) and magnitude poorly. The direct delta model failed and is not used.

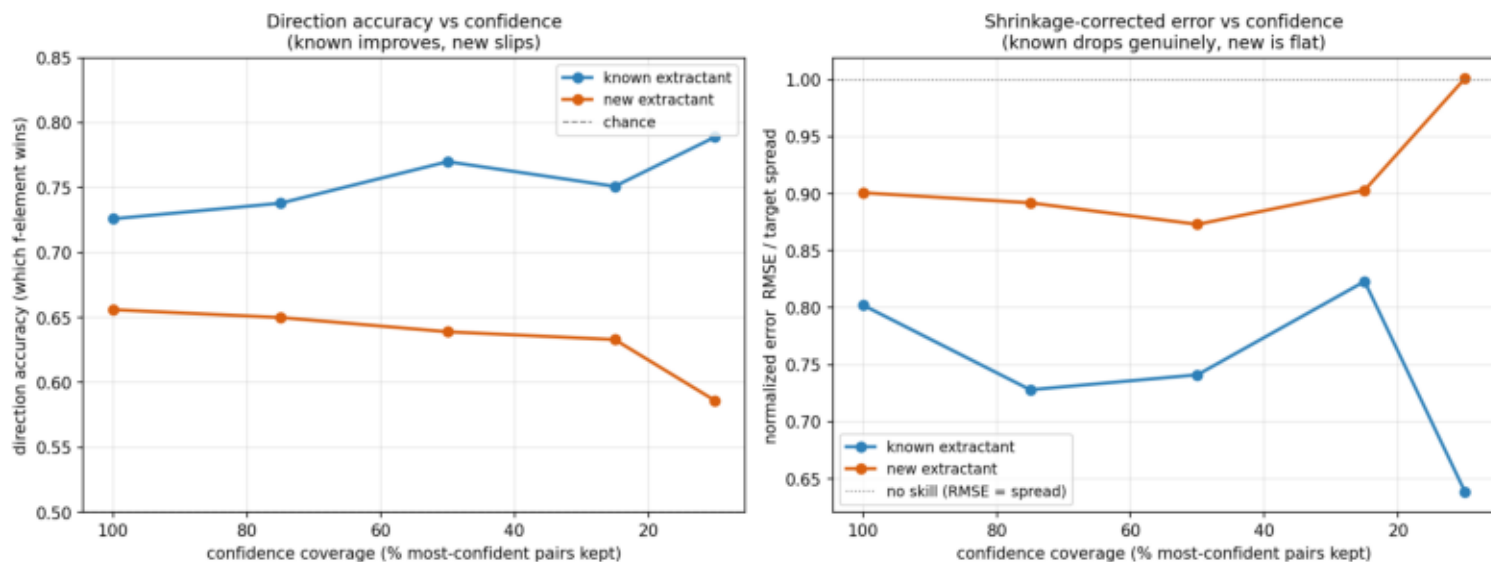


Figure 7. Separation prediction improves with confidence only for a known extractant (direction 0.73 to 0.79; normalized error beats a random-ranking benchmark, adversarially verified). For a new extractant confidence is not net-useful.

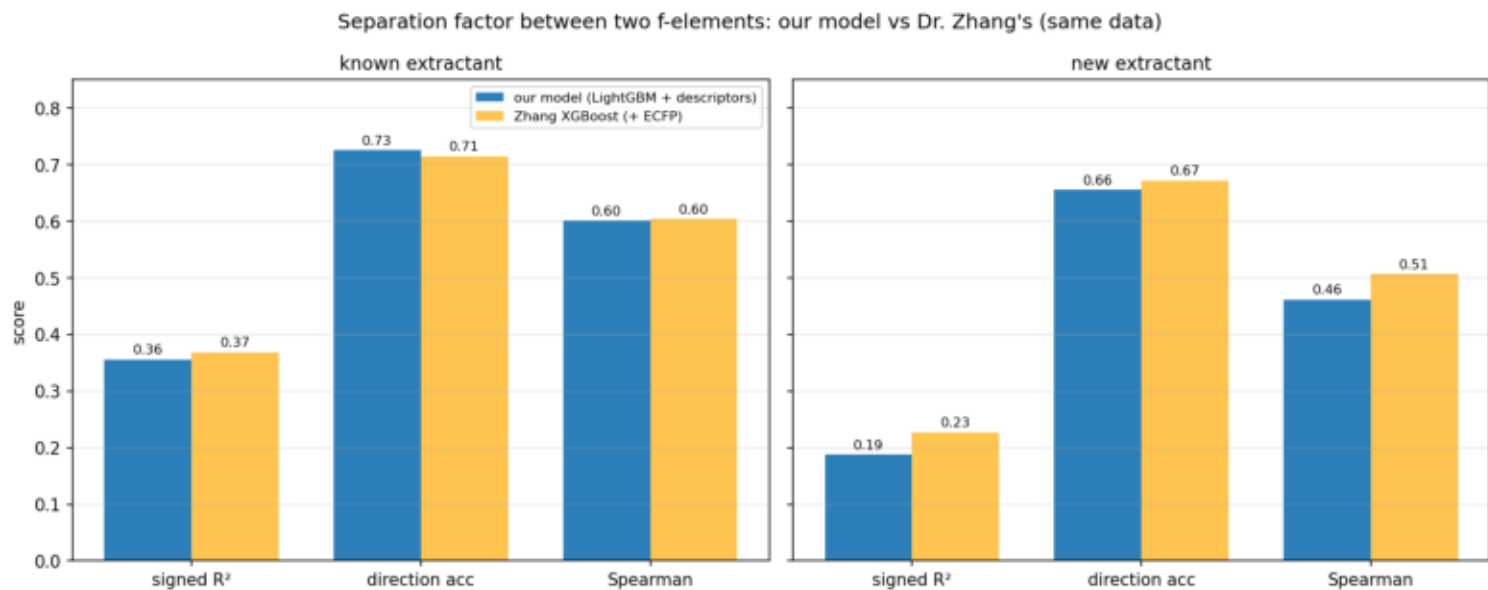


Figure 8. Our model vs Dr. Zhang's XGBoost on the separation factor (same data): they tie at full coverage, but only our model carries a confidence layer, so only ours has the confidence-filtered operating point (direction 0.789).

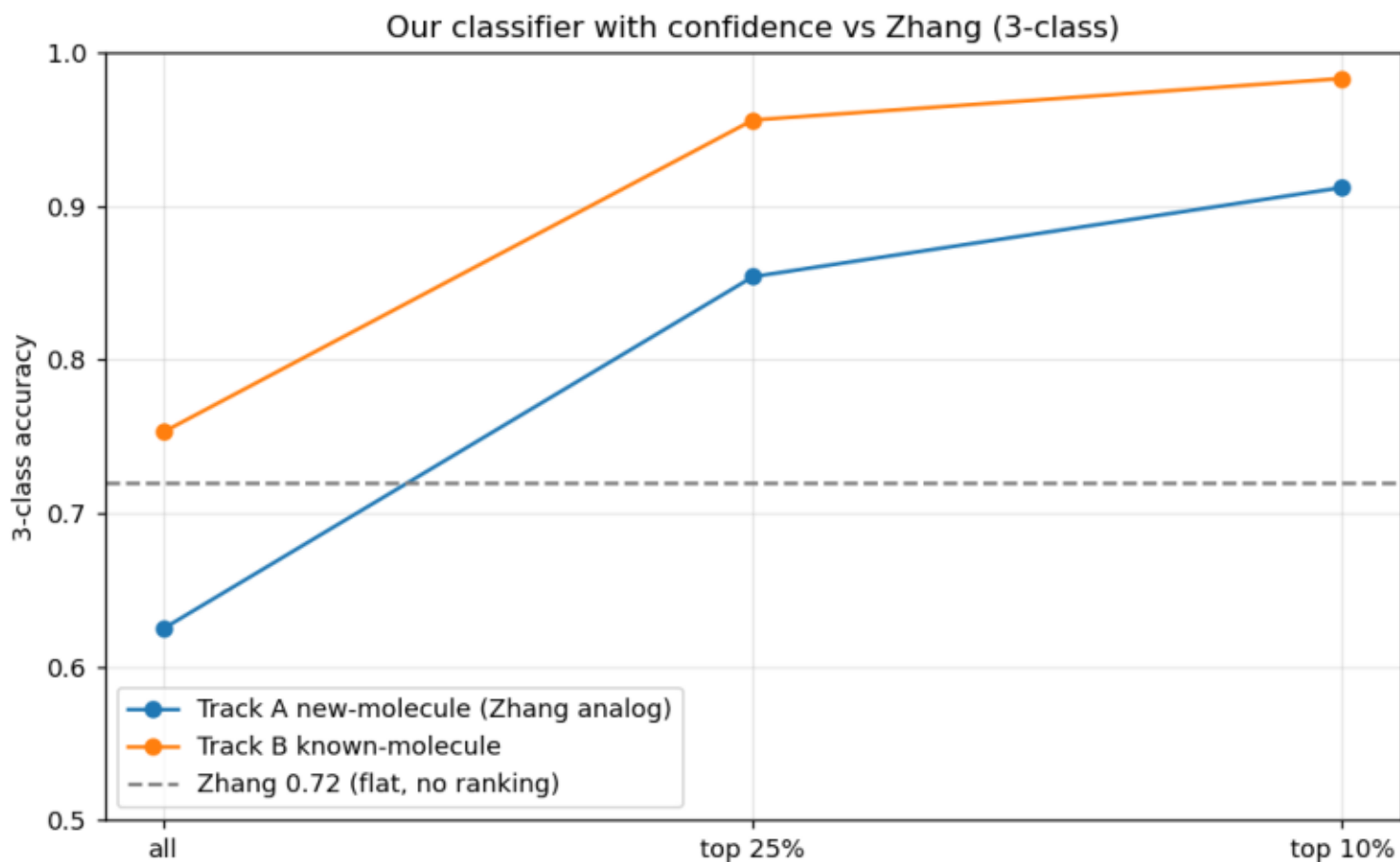


Figure 9. Comparison with Dr. Zhang on logD (same data): about equal in accuracy; our addition is the calibrated, confidence-aware prediction, not higher raw accuracy.

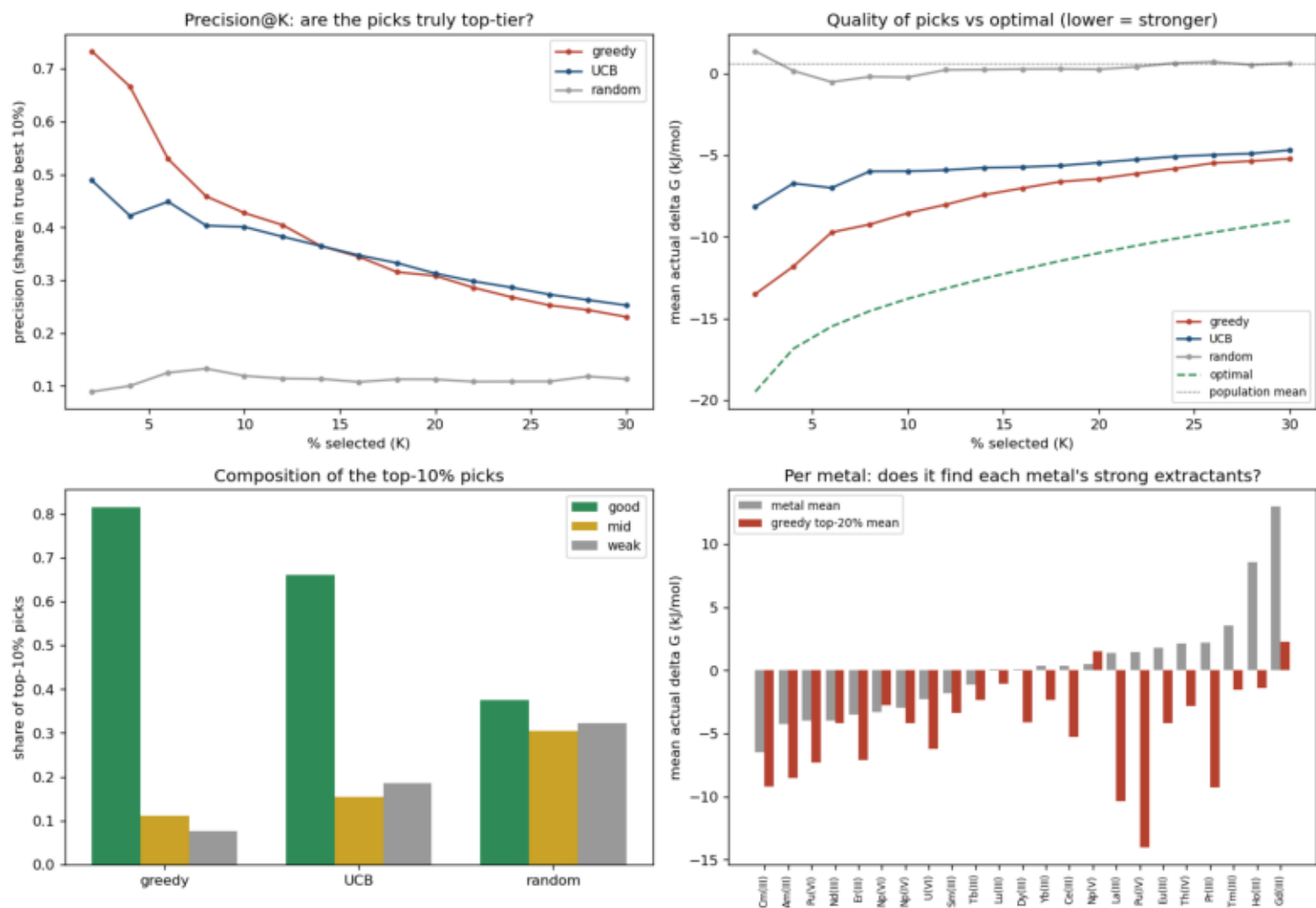


Figure 10. Experiment prioritization: ranking candidates by prediction concentrates strong extractants near the top (81 percent of top picks in the strongest tertile, 3.6x random).

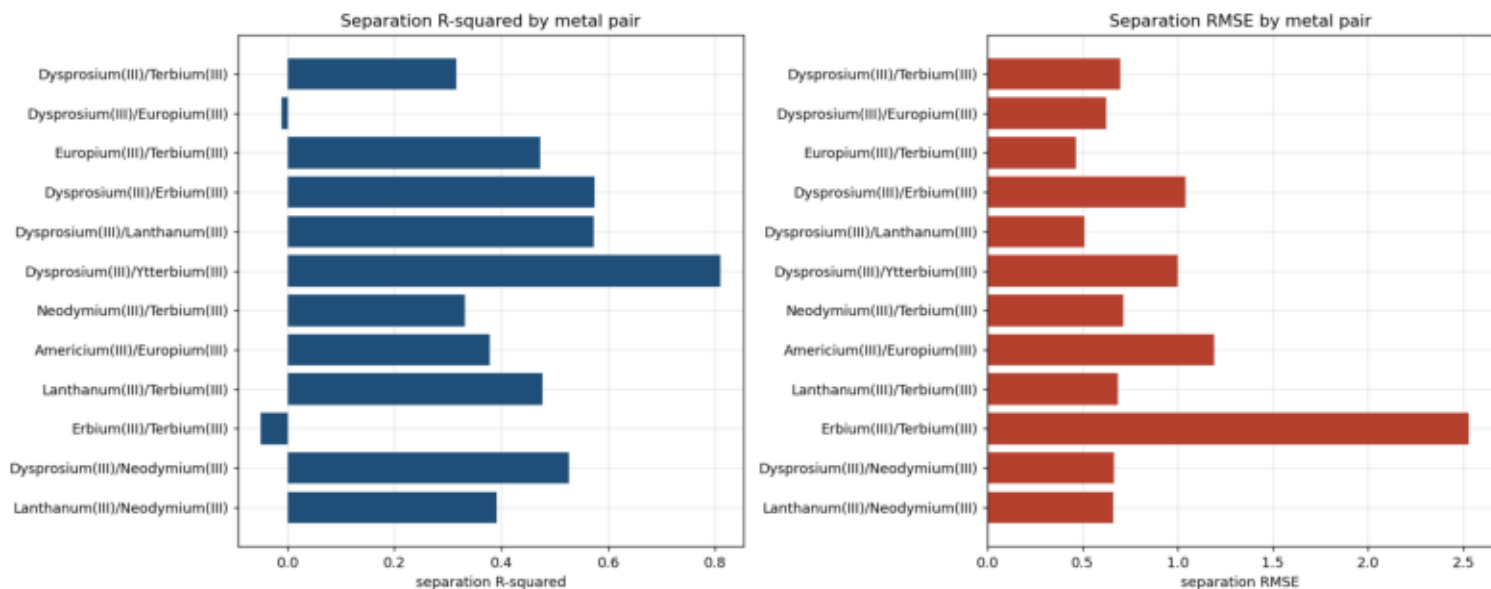


Figure 11. Per-pair separation accuracy is highly uneven across f-element pairs: large-gap pairs predict well, adjacent rare earths do not.